

2026 Net Zero Tech — Summary Table of Project Descriptions

Track: Sustainability — (1) Carbon Management & Climate Tech (Carbon Removal / Negative Emissions, MRV). Format: Arial 12, 18 pt line spacing, A4. Fields in 【 】 are placeholders.

No.	(assigned by organizer)
Work Title	LeafCarbon: low-cost, non-destructive estimation of plant carbon from images
School	【School】
Department	【Computer Science】 / 【Materials Science & Engineering】
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1. Introduction of Team

We are a cross-disciplinary team of Computer Science and Materials Science students. The Computer Science side works on the image machine-learning model, the confidence-based routing logic, and the demo system. The Materials Science side works on the oven dry-weight calibration, the imaging standardization, and the measurement procedure. Between the two sides we cover both parts that a measurement needs: the model and the reference values it is checked against.

2. Creation Motive — *problem-solving and technical value*

For biological carbon removal (afforestation, soil, restoration, microalgae), the difficulty is often in the measurement: knowing how much carbon was captured, accurately and at low cost. This is the MRV step (Monitoring, Reporting, Verification). The standard way to measure per-plant biomass is destructive harvest, oven-drying to constant weight, then weighing each plant. It destroys the plant, has to be done plant by plant, and is too expensive to apply to many smallholders or large areas. Because of this, operators that do capture carbon often cannot earn credits: the amount is hard to measure affordably and its permanence is hard to prove. Taiwan's carbon fee (NT\$300 per t CO₂e from 2025) and the national target of +10 Mt CO₂e of agricultural sinks by 2040 both depend on measurement that is currently expensive to scale. Image-based MRV for biological carbon has so far received relatively little attention. LeafCarbon is a low-cost way to do this measurement.

3. Research Process

(1) We built our own dataset: lettuce we grew ourselves, photographed day by day, paired with dry weights and on-site environment records. (2) We segmented the plant out of the low-cost images (green-canopy extraction) and computed structure features (canopy area, plant height, canopy density). (3) We added environmental history (cumulative light, growing-degree-days). (4) We

regressed per-plant dry biomass with a Random Forest and validated it with leave-one-day-out cross-validation. (5) We anchored the model to ground truth with a small set of oven-dried dry weights. Carbon is then taken as dry weight $\times 0.47$, a widely used constant, so elemental analysis is not required. Each estimate carries a confidence interval, so only uncertain samples are sent to the lab.

4. Brief of Work — *creativity / technical content and feasibility*

LeafCarbon is a measurement setup rather than an app: standardized low-cost imaging, an AI estimator, and a small calibration set. The carbon fraction of plant dry matter is close to constant (about 47%; IPCC 2006 default 0.47 tC/t; Ma et al. 2018 report 45.0–47.9% across 4,318 species). The quantity that actually varies is biomass, so carbon = biomass $\times 0.47$. The image estimates biomass, and the small set of oven-dried weights calibrates the image-to-biomass relationship; carbon then follows from the 0.47 constant, with no elemental analysis. Estimating biomass from leaf structure is an established approach (for example NDVI and SPAD), so the method builds on validated work and our effort goes into the integration. What we add is a low-cost, non-destructive measurement tool set up for carbon projects. Lettuce plays the role of a fast validation crop here: its roughly 35-day cycle lets us tune the image-to-biomass relationship and prove the workflow quickly, and the same method is built to transfer to durable pools such as forests, soil, and biochar.

5. Expected Benefit — *specific expected efficiency, output value, data*

On our own lettuce image set (569 photos over days 7–35), the per-plant biomass model is trained and validated on the 72 single-plant samples that have individual dry weights. Using side-view structure features together with cumulative environment, it reaches $R^2 \approx 0.64$ and $RMSE \approx 0.43$ g under leave-one-day-out cross-validation. This is a measured result on our data, with the model fully specified (image and environment $\rightarrow$ biomass $\rightarrow$ carbon). On cost and speed: the traditional way to get one plant's biomass is destructive harvest, oven-drying, then weighing, done plant by plant and never repeatable; LeafCarbon estimates it from a photo in seconds, without destroying the plant, and the same plant can be measured again later. At project scale, digital MRV has been reported to reduce per-farm verification cost by roughly 70–90% (Ecosystem Marketplace, 2025 State of the Voluntary Carbon Market). This would let afforestation, soil-carbon, and restoration projects, and smallholders, afford verification, and it aligns with Taiwan's carbon fee and the 2040 agricultural-sink target.

6. Project Introduction Video (≤ 5 min, MPEG-4, English, download enabled)

Cloud link: 【paste URL with download ON】. Flow: the problem (measuring carbon is too expensive to scale) $\rightarrow$ what we built (a low-cost imaging measurement setup) $\rightarrow$ footage (the capture rig and the oven dry-weight calibration) $\rightarrow$ AI demo (photo and environment $\rightarrow$ biomass and carbon, with a confidence interval) $\rightarrow$ measured results and a cost comparison.